

Selection Sort

15	28	17	12	18	9	6
0	1	2	3	4	5	6

→ find smallest → swap

6	28	17	12	18	9	15
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→ swap 9 with 28

6	9	17	12	18	28	15
---	---	----	----	----	----	----

swap 12 with 17

6	9	12	17	18	28	15
---	---	----	----	----	----	----

swap 15 with 17

6	9	12	15	18	28	17
---	---	----	----	----	----	----

Swap 17 with 18

6	9	12	15	17	28	18
---	---	----	----	----	----	----

Swap 18 with 28

6	9	12	15	17	18	28
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Pseudocode: and Time Complexity

int Sel_Sort (int arr[])

{

int

temp; ^①

int min_index; ^①

for (int i = 0; i < arr.length - 1; i++)

{ $\hookrightarrow 1 + n + n - 1 = 2n$

min_index = i; $(n-1)$

for (j = i + 1; j < arr.length; j++)
 $\hookrightarrow (n-1) + (x+1) + x$

{

if ($arr[\text{min_index}] > arr[j]$)
↳ (3x)

{

$\text{min_index} = j$;

}

}

// swap $arr[i]$ & $arr[\text{min_index}]$

$\text{temp} = arr[i]$;

$arr[i] = arr[\text{min_index}]$;
↳ 2(n-1)

$arr[\text{min_index}] = \text{temp}$;
↳ 2(n-1)

}

return arr ;

}

$$= 1 + 1 + 2n + n-1 + n-1 + x + 1 + x + 3x + x + 2(n-1) + 3(n-1) + 2(n-1) + 1$$

$$= \cancel{1} + \cancel{1} + 2n + \cancel{n-1} + \cancel{n-1} + \cancel{x} + \cancel{1} + \cancel{x} + 3\cancel{x} + \cancel{x} + 2\cancel{n-1} + 3\cancel{n-1} + 2\cancel{n-1} + 1$$

$$= 6x + 11n - 5$$

where

$$x = 1 + 2 + 3 + \dots + (n-1)$$

In A.P

$$1 + 2 + 3 + 4 + \dots + k = \boxed{\frac{k(k+1)}{2}}$$

$$x = \frac{(n-1)(n-1+1)}{2}$$

$$\boxed{x = \frac{n(n-1)}{2}}$$

$$= 6x + 11n - 5$$

$$= 11n + 6x - 5$$

$$= 11n + \frac{6n(n-1)}{2} - 5$$

$$= 11n + \frac{6n^2}{2} - \frac{6n}{2} - 5$$

$$= 11n + \underbrace{3n^2 - 3n} - 5$$

higher order term is n^2 .

$\therefore$ The time complexity is

$$\boxed{O(n^2)}$$
